

Mini Project Report

SRM INSTITUTE OF SCIENCE AND TECHNOLOGY
College of Engineering and Technology
Department of Electronics and Communication Engineering

21ECC211L / Devices and Digital IC Lab

III Semester, 2024- 25(Odd Semester)

Lab code & Title : 21ECC211L / Devices and Digital IC lab

Year & Semester : 2nd Year, Sem 3

Project Title : 2 bit binary adder using full adder

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Reg. No & Name			
Mark split up	RA2311004010363 Abel Baby	RA2311004010364 Akshaya M	RA2311004010365 Saksham Sinha
Novelty of the project (1 mark)	0	1	1
Level of understanding of the Project (2 marks)	1	1.5	2
Contribution of the Project Work (1 mark)	1	1	1
Project Report (1 Mark)	0.5	1	1
Total (5 Marks)	2.5	4.5	5

REPORT VERIFICATION

Date: 10/10/2024 .

Signature of Lab Supervisor



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OBJECTIVE

The primary objective of this project is to design and implement a 2-bit binary adder utilizing full adders, demonstrating fundamental concepts in digital electronics and binary arithmetic. By constructing this circuit, we aim to enhance our understanding of binary addition and the operation of full adders in digital systems.

ABSTRACT

This project explores the design, implementation, and functionality of a 2-bit binary adder constructed using full adders. Binary addition is a crucial operation in computing and digital circuits, serving as the foundation for more complex arithmetic tasks. This report details the hardware requirements, circuit design, and testing results of the 2-bit binary adder. The successful implementation demonstrates the correct operation of the adder across all possible input combinations.

providing practical insights into binary arithmetic and digital logic design.

INTRODUCTION

In modern computing, the binary numeral system is fundamental, representing all data and instructions in digital format. Binary addition is one of the simplest yet most important operations performed by computers. A binary adder is a digital circuit that performs this arithmetic operation, and it can be implemented using full adders.

A full adder is a combinational circuit that adds three binary bits: two significant bits and a carry bit from a previous addition. The outputs of the full adder are a sum bit and a carry-out bit. This project focuses on constructing a 2-bit binary adder that can add two 2-bit binary numbers, producing a 2-bit sum and a carry-out if the result exceeds the binary representation of 3 (i.e., 11 in binary).

Importance of the Project

Understanding binary addition and the operation of full adders is crucial in digital electronics. This knowledge serves as a building block for more complex operations, such as subtraction, multiplication, and division in digital circuits. Additionally, exploring binary adders provides practical experience in circuit design and implementation, valuable skills for students and professionals in the field of electronics and computer engineering.

SOFTWARE & HARDWARE REQUIREMENTS

/DESCRIPTION

To construct the 2-bit binary adder, the following components are required:

1. Full Adder IC (e.g., 74LS83):

- The 74LS83 is a 4-bit binary adder IC that can accommodate our 2-bit addition needs. It has multiple inputs for binary digits and produces a sum and carry-out, making it ideal for our project.

2. Breadboard:

- A breadboard is essential for assembling the circuit without soldering. It provides flexibility for modifications and testing, allowing easy reconfiguration of the circuit.

3. Connecting Wires:

- Wires are necessary to connect various components on the breadboard, facilitating signal flow between the full adders, power supply, and output devices.

4. Power Supply (5V):

- A regulated 5V power supply is needed for the proper operation of the full adder IC. Consistent voltage levels are crucial for reliable circuit performance.

5. LEDs:

- LEDs will visually indicate the output sum and carry results. Each output bit will connect to an LED, enabling easy observation of the adder's operation.

6. Resistors (220 Ohm):

- Resistors are required to limit the current flowing through the LEDs, protecting them from damage and ensuring proper functionality.

7. Switches:

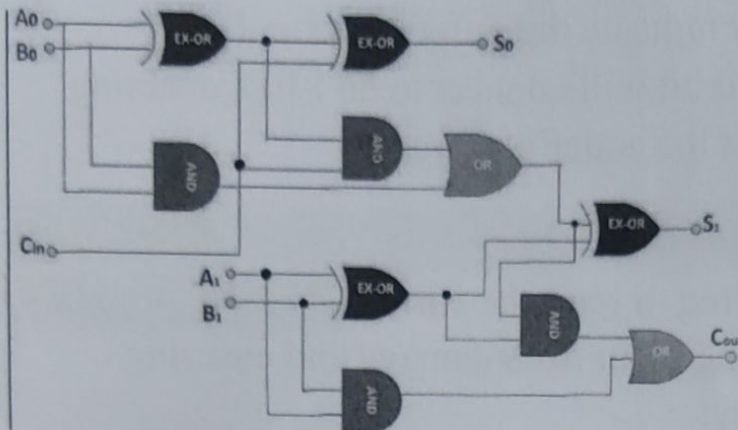
- Switches will serve as input devices for entering the binary numbers. Each switch will represent one bit of the binary inputs.

8. Multimeter:

- A multimeter will assist in testing voltage levels and confirming that the circuit operates correctly. It will be particularly useful in troubleshooting any issues that arise during testing.

CIRCUIT DIAGRAM AND DESCRIPTION

The circuit diagram for the 2-bit binary adder is illustrated below:



The circuit consists of two full adders connected in series to perform the binary addition. The first full adder handles the least significant bits (LSBs) of the two binary numbers, along with an initial carry input (set to 0). The outputs of this first full adder include:

- Sum Output: This represents the LSB of the final result.
- Carry Output: This is forwarded to the second full adder.

The second full adder adds the most significant bits (MSBs) of the two binary numbers, incorporating the carry output from the first full adder. The outputs of the second full adder yield:

- Sum Output: This represents the next significant bit of the result.
- Carry Output: This provides the final carry-out if the result exceeds the 2-bit binary limit.

This arrangement enables the addition of two 2-bit binary numbers, generating a 2-bit sum and a carry-out.

RESULTS AND DISCUSSION

Upon implementing and testing the 2-bit binary adder, the following results were observed for various binary inputs:

INPUTS					OUTPUTS		
A0	A1	B0	B1	Ci	S0	S1	Co
0	0	0	0	0	0	0	0
1	0	0	0	0	1	0	0
0	1	0	0	0	0	1	0
1	0	1	0	0	0	1	0
0	1	0	1	0	0	0	1
0	1	0	0	1	1	1	0
1	0	1	0	1	1	1	0
1	1	1	1	1	1	1	1

Analysis of Results

The results confirm that the adder operates correctly across all possible 2-bit binary inputs. The outputs align with expected results based on binary addition rules:

1. Sum of 00 and 00 yields a sum of 00 with no carry.
2. Sum of 01 and 01 produces a sum of 10, indicating that $1 + 1$ in binary results in 0 (sum) and a carry of 1.
3. Sum of 11 and 11 results in a carry, demonstrating the adder's capability to handle overflow, where $1 + 1 + 1$ (from the previous carry) yields 1 with a carry-out.

The circuit was tested multiple times to ensure reliability and accuracy, and in all instances, it produced the expected outputs. This confirms that the full adders functioned correctly.

Limitations

While the project successfully demonstrates the operation of a 2-bit binary adder, it has limitations. The current design can only handle 2-bit binary numbers, which restricts its utility in larger computations. Future iterations could focus on expanding the adder's capability to handle larger binary numbers, potentially through the use of more full adders in series.

CONCLUSION

The 2-bit binary adder constructed using full adders effectively illustrates the principles of binary addition in digital circuits. This project highlights the operational capabilities of full adders and provides hands-on experience in circuit design and implementation. The successful addition of 2-bit binary numbers confirms the accuracy of the design and its potential for further applications in more complex arithmetic operations.

Future Work:

Future projects could explore the development of larger binary adders capable of adding multi-bit binary numbers by cascading multiple full adders. Additionally, integrating features such as carry-lookahead could improve performance and efficiency in larger adder designs. Exploring alternative arithmetic operations, like subtraction or multiplication, could also expand the understanding of digital arithmetic and its applications.

REFERENCES

- Wikipedia contributors. (2023). Full adder. Wikipedia, The Free Encyclopedia. Retrieved from [Wikipedia](#).
- IEEE. (2022). IEEE Standards for Digital Circuits. IEEE Standards Association.